TAKSH VADALIYA

AI APPLICATION ENGINEER / PRODUCT BUILDER

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LinkedIn | GitHub | Portfolio

FOCUS

AI application engineering

Retrieval and RAG systems

Evaluation and data quality

Voice and computer vision prototypes

TECHNICAL TOOLKIT

Languages: Python, C, TypeScript

Backend: FastAPI, REST, async services, PostgreSQL

AI: RAG, hybrid retrieval, embeddings, chunking, semantic caching, agent orchestration

Search: Qdrant, FAISS

Voice: STT / LLM / TTS, turn-taking, barge-in

Vision: RTMPose, human parsing, image geometry

Product: API design, auth, billing, deployment

EDUCATION

B.Sc. Information Technology
GTU-ITR, Gujarat | 2023-2026
CGPA: 8.5 / 10

RECOGNITION

GTU Ventures / Gujarat Fellowship
Assest selected for incubation, March 2026

Devang Mehta Hackathon 2025
Team lead; built and demoed an end-to-end platform

Smart India Hackathon 2024
Qualified for state round

LANGUAGES

English · Hindi · Gujarati

PROFILE

AI application engineer and founder building software around AI models, with hands-on work across retrieval, evaluation, backend APIs, and real-time voice. Took a voice-AI concept to a public product and built substantial prototypes in retrieval and computer vision. Seeking AI Application Engineer roles focused on shipping useful, dependable software.

EXPERIENCE

AI Algo Builder | Jainam Broking | Jun 2026-Present
Researching and iterating on systematic trading strategies using historical market data, news signals, and price patterns. Work remains internal.

Founder & AI Engineer | SolidExposure | Mar-Jun 2026
Designed and built a public voice-AI product with multilingual conversation, interruption handling, backend services, and product flows for accounts and credits. Led early product discovery and commercial exploration.

Founder | Assest | 2026-Present
Designed and built a company-knowledge prototype spanning ingestion, hybrid retrieval, and access-control concepts; selected for GTU Ventures / Gujarat Fellowship.

SELECTED PROJECTS

Assest - Agentic company knowledge prototype Python, FastAPI, PostgreSQL, Qdrant
Hybrid dense and lexical retrieval, reciprocal rank fusion, semantic caching, and access checks. Repository.

Qual - Evaluation-first RAG Python, FAISS
Market-intelligence pipeline with a four-part rubric for groundedness, completeness, relevancy, and numeric-risk. Repository.

Body-aware shopping compatibility Computer-vision prototype
Colab exploration using RTMPose landmarks, human parsing, and image geometry. Estimates are unvalidated; product ranking is future work. Case study.

DDAS - Duplicate detection in C Systems fundamentals
Custom hashing and nested file-tree scanning. Under active rebuilding; no test suite or benchmark is claimed. Repository.

TRAINING

Additional learning: Harvard/edX CS50 Python · Microsoft AI/ML bootcamp · Anthropic Claude 101 · Mastercard Cybersecurity · Cisco Endpoint Security